

PicoCTF 2023 - Reverse

Reverse

Reverse Engineering

Medium

100 pts

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Try reversing this file? Can ya?

I forgot the password to this file [↗]. Please find it for me?

In this challenge, we're asked to obtain a password from a file

PicoCTF 2023 – Reverse Reverse Engineering

```
└─$ file ret  
ret: ELF 64-bit LSB pie executable, x86-64,  
86-64.so.2, BuildID[sha1]=6c43c1779ecbf9a8d
```

After downloading the file, we determine that this
is an executable binary

PicoCTF 2023 – Reverse String Enumeration

```
└─$ strings ./ret  
/lib64/ld-linux-x86-64.so.2  
libc.so.6  
__isoc99_scanf  
puts
```

Since this is a binary file, we could get some further understanding of it by extracting the text strings from it

PicoCTF 2023 – Reverse String Enumeration

```
└─$ strings ./ret  
/lib64/ld-linux-x86-64.so.2  
libc.so.6  
__isoc99_scanf  
puts
```

This is a common initial step in software reverse engineering, because text strings can contain sensitive data, like passwords

PicoCTF 2023 – Reverse String Enumeration

```
└─$ strings ./ret | grep pico  
picoCTF{H  
Password correct, please see flag: picoCTF{3lf_r3v3rs1ng_succ3ssful_p4ssw0rd}
```

By searching the strings for the picoCTF flag pattern, we discover the flag